

Technical Note

CROSSLINKING FOR PAPER, TEXTILES AND POLYVINYL ALCOHOLS

PAPER

Dorf Ketal Tyzor® titanates crosslink PVA and starch in adhesive formulations to improve water resistance¹⁻⁴ of Kraft paper laminates and corrugated paperboard. The titanates also are used in sizing formulations to reduce absorption of oils, water and inks, and to improve wet strength.⁵⁻¹¹

Titanates are added to paper coatings to improve coating adhesion and to improve sizing and tear-strength properties.¹²⁻¹⁶ Paper pulp treated with aqueous siloxane sizing emulsions containing Tyzor® TE, AA-75 or GBA improve water repellency and anti-adhesion properties.¹⁷

TEXTILES

Dorf Ketal Tyzor® titanates can be used by themselves or in combination with waxes or resins such as vinyl ether polymers, acrylic copolymers and polysiloxanes to treat natural and synthetic materials. The titanates improve water repellency¹⁸⁻²⁶ and dye retention,²⁷ as well as delustering, slipping,²⁸ and soil resistance²⁹ in these formulations.

POLYVINYL ALCOHOL INSOLUBILIZER

Dorf Ketal Tyzor® AA, AA-75, AA-65, AA-105, GBA, and GBO organic titanates (called AA-Series) are effective insolubilizers for polyvinyl alcohols such as DuPont Elvanol® when properly pretreated prior to mixing.

To prevent rapid gelation, a stable aqueous titanate solution must be prepared first by mixing the Tyzor® AA-Series titanate with equal parts of 3% hydrochloric acid, heating to 85 °C (185 °F) and allowing the mixture to cool. Solutions prepared in this manner can be diluted and mixed with PVA solutions to yield mixtures that show no increase in gel structure for extended periods at room temperature.

If these mixtures contain more than 5% polyvinyl alcohol, there is a slight initial rise in viscosity when the Tyzor® titanate is added but effectiveness is not impaired. As little as 5% Tyzor® AA-Series titanate, based on the dry weight of the PVA, can be used. The optimum amount and required degree of stability should be determined by the user.

Dorf Ketal Tyzor® TE and LA organic titanates also may be used as insolubilizers. These products do not require acid treatment before use although a small amount of ammonia may be required to maintain alkalinity.

PVA insolubilization by organic titanates occurs at room temperature, and more rapidly at elevated temperatures. Films cast from solutions containing Elvanol® polyvinyl alcohol, glycerin and an organic titanate have improved water resistance after standing overnight, but a more pronounced and immediate effect will result if such films are briefly heated to 80°C (176 °F).

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